VISION OF THE DEPARTMENT

To nurture competent, innovative, ethical, and socially responsible computer science professionals with research aptitude and lifelong learning skills to address emerging technological and societal challenges.

MISSION OF THE DEPARTMENT

M1: Provide quality education through a contemporary curriculum integrating computing fundamentals, emerging technologies, industry practices, and multidisciplinary learning.

M2: Foster innovation, research, entrepreneurship, and problem-solving abilities to develop sustainable technological solutions for societal and industrial needs.

M3: Create a learner-centric environment that promotes professional ethics, leadership, teamwork, communication skills, and lifelong learning.

M4: Strengthen industry-academia collaboration and encourage continuous professional development to meet global technological challenges.

Program Educational Objectives (PEOs)

PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

PEO1: Establish successful careers in industry, entrepreneurship, research, or higher education by applying strong foundations in computer science and engineering principles.

PEO2: Develop innovative, ethical, and sustainable solutions to real-world problems through professional competence, multidisciplinary collaboration, and lifelong learning.

PEO3: Demonstrate leadership, communication, and managerial skills while contributing responsibly to society and the advancement of technology.

PROGRAM OUTCOMES

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering Fundamentals and an engineering specialization for the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyses complex Engineering problems reaching substantiated conclusions using first principles of mathematics,Natural sciences and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling to complex engineering activities, with an understanding of the limitations.

PO6: The engineer and world: Apply reasoning informed by the contextual knowledge to assess Societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8: Individual and Collaborative team work Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10: Project management and finance: Demonstrate knowledge and understanding of the Engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Knowledge and Attitude Profile (Washington Accord Knowledge)

WK 1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK 2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK 3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK 4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK 5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK 6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK 7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK 8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK 9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc., with mutual understanding and respect, and of inclusive attitudes.

Program Specific Outcomes (PSOs)-UG

PSO1: Design, develop, test, and deploy efficient software systems using programming, algorithms, data structures, and modern software engineering practices.

PSO2: Apply emerging technologies such as Artificial Intelligence, Data Science, Cyber Security, Cloud Computing, Internet of Things, and Machine Learning to solve complex engineering problems.

Scheme of Teaching and Examination for 160 Credits (2026 Scheme
Department of Computer Science and Engineering

1st Semester B.E. Academic year 202

Sl. No.	Course Code		Course Title	Teach. Dept.	Credits	Teaching Hrs. / Semester			
						L	T	P	S
1	ASC	26MA101	Discrete Mathematical Foundations	MA	4	42	28	-	50
2	ESC	26CS102	Python Programming	CS	3	42	-	-	48
3	PSC	26CS103	Web Essentials	CS	3	42	-	-	48
4	ESC	26HS104	Computer Fundamentals	CS	4	56	-	-	64
5	ESCL	26CS105	Python Programming Foundations Lab	CS	1	-	-	28	2
6	PSCL	26CS106	Web Essentials Lab	CS	1	-	-	28	2
7	HSC	26HS107	Industrial Economics	HS	2	28	-	-	32
8	HSC	26HS108	Communication Skills	HS	1	-	-	28	02
9	HSC	26IK109	Indian Knowledge System	HS	1	14	-	-	16
10	HSC	26HS110	Constitution of India	Any	PP	0	0	28	2
				Total	**20**	**22**	**28**	**11**	**26**

LEGEND: XX: Department Code, L: Lecture Hours, T: Tutorial Hours, P: Practical / Drawing Hours, S: Student Academic Activity Hours, CIE: Continuous Internal Evaluation, SEE: Semester End Examination

Credit Distribution: ASC: Applied Science Course (4), ESC: Engg. Science Course (6), ESCL: Engg. Science Lab. Course (1), PSC: Program Specific

Course (3), PSCL: Program Specific Core Laboratory (1), HSC: Humanity and Social Science Course (3), PBL-Project base learning (2)
CIE Pattern: IA: Internal Assessment, ABL: Activity Based Learning**Non-integrated Course:** Theory IA (40 Marks), ABL (10 Marks) [**SEE**: Min. 40% in Theory IA & Min. 40% in ABL]

Syllabus for the Academic Year 2026-27

Department: Computer Science and Engineering			Semester:	I
Subject: Discrete Mathematical Foundations (Common to AI&ML,CS(AI&ML), CS,CR,DS, IS)				
Subject Code:	26MA101	L – T – P - S:	42-28-0-50	
			Credits:	4

Sl. No	Course Objectives
1	Formulate Mathematical Induction and Logic.
2	Analyze and solve engineering problems by applying LCM, GCD, Relations, Derangements and Pigeonhole Principle.
3	Demonstrate the use of Number Theory and Logic to solve the engineering problems.
4	Apply the concepts of Logic, Relations and Functions in project related work.

Module	Description	Hrs
I	**Sets:** Set operations, Inclusion-Exclusion Principle, Cartesian Product, Countable and Uncountable Sets, Cardinality of Number Systems. **Number Theory:** Properties of integers, Division algorithm, Divisibility rules, Prime numbers, The greatest common divisor(GCD), Least common factors(LCM), Prime Factorization, Modular arithmetic, Euclid's algorithm and Applications- Problems. Using modern tool to verify prime, composite numbers and compute factorization, LCM, GCD. (Text book 1: 1.2 : Text book 2:3.1, 8.1, 5.1, 4.3,4.4,4.5)	9
II	**Relations & Functions:** Relations, Properties of relation, Computer Recognition: Zero-One Matrices and Digraphs,Equivalence Relations, Partial Ordered Relation and Hasse Diagrams. **Types of Functions:** Functions, Composition, Inverse Functions- Problems. Solving and visualizing relations and functions using modern tool. (Text book 1:4.2 : Text book 2:5.1,7.2, 7.4,7.3,5.2,5.3)	9
III	**Mathematical Induction:** The well-defined principle, Weak Mathematical Induction, Strong Mathematical Induction. **Combinatorics:** Counting Principles, Rule of Sum and Product, Permutations, Combinations, Binomial Theorem, Stars & Bars, Derangements, Pigeonhole Principle - Problems. Using modern tool to solve and analyze combinatorics and mathematical induction. (Text book 1: 3.1,3.2,3.3, : Text book 2:4.1)	8
IV	**Propositional Logic:** Introduction, Propositions, Compound Propositions, Basic Logical Connectives, Truth Tables. Tautology, Contradiction and Contingency. Logical Equivalence. Laws of Logic - Problems. Solving and analyzing propositional	8

V	logic using modern computational tools. (Text book 1:2.1 : Text book 2:2.1)	8

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Discrete and Combinatorial Mathematics	Ralph P. Grimaldi	5th Edition, Pearson Education, 2012, ISBN-978-0201726343
2	Discrete Mathe matics and its Applications	Kenneth H. Rosen	8th Edition, 2021, ISBN-13-978-939072735

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Discrete Mathematical Structures with Applications to Computer Science	J.P. Tremblay and R. Manohar	1st Edition 2017, Tata – McGraw Hill, ISBN-13:978-0074631133 .
2	Discrete Mathematical Structures	Bernard Kolman, Robert C. Bussy, Sharan Cutler Ross	6th Edition, PHI learning private limited, ISBN-978-81-203-3689-6

Course Outcomes:

Course outcome	Description
CO1	Construct valid logical arguments and formal proofs using rules of inference and predicate logic.
CO2	Apply set operations, divisibility rules, and number theory concepts to model
CO3	al techniques
CO4	lences using

Course Articulation Matrix:

CO / PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1													
CO2													
CO3													
CO4													
CO													

CO1	3	2									3		
CO2	2	3		2							3		
CO3	2	3									3		
CO4			3	2							2		
CO5	2	2	2								2		

Department: Computer Science and Engineering			Semester:	I
Subject: Python Programming Foundations				
Subject Code:	26CS102	L – T – P - S:	42-0-0-48	
			Credits:	3

Sl. No	Course Objectives	
Module	Description	Hrs
2 I	Develop Python programs using variables, operators, control flow statements, loops, functions, and data structures.	
3	Apply problem-solving techniques to design and implement efficient Python solutions.	
4 II	Utilize object-oriented programming concepts, recursion, file handling, and exception handling in real-world applications.	

Module	Description	Hrs
I	**Introduction to Programming and Python Fundamentals** Python basics, interpreters, and environment setup ; Variables, data types, and arithmetic operators; Input/Output operations; Computational thinking and problem-solving basics	
II	**Control Flow and Iterative Problem Solving** Conditional branching (if, elif, else); Looping structures (for, while); Loop control statements (break, continue, pass); Nested loops and logical problem solving	
III	**Functions, Strings and Collections** Function definition, parameters, and scope; String manipulation, indexing, and slicing; Lists, Tuples, and Dictionaries; Set operations and data organization	
IV	**Data Structures and Recursion**	

	Advanced list and dictionary operations; Recursion fundamentals and algorithm design; Base cases and recursive stack; Iteration vs. Recursion	
V	**Object-Oriented Programming and File Handling** Classes, objects, and method definitions; Inheritance, polymorphism, and encapsulation; File I/O operations (read, write, append); Exception handling and robust code management	

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Think Python: How to Think Like a Computer Scientist	Allen B. Downey	3rd Edition, 2024
2	Automate the Boring Stuff with Python: Practical Programming for Total Beginners	Al Sweigart	2nd Edition, 2019

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Python Crash Course	Eric Matthes	3rd Edition, 2023
2	Head First Python	Paul Barry	2nd Edition, 2016
3	Learning Python	Mark Lutz	5th Edition, 2013

Course Outcomes:

Course outcome	Descriptions
CO4	**Implement** recursive solutions and object-oriented programs using classes and OOP principles.
CO1	**Explain** Python applications, fundamentals, and problem-solving algorithms, and Python syntax.
CO5	**Build** programs using file I/O and exception handling techniques.
CO2	**Develop** Python programs using variables, operators, conditional statements, loops, and functions.
CO3	**Apply** strings, lists, tuples, dictionaries, and sets to solve computational problems.
CO4	**Implement** recursive solutions and object-oriented programs

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3				2			1			1	3	2
CO2	3	2	2		3							3	2
CO3	2	3	2		3							3	2
CO4	2	3	3		2							3	2
CO5	2	2	3		3							2	3

Syllabus for the Academic Year 2026-27

Department: Computer Science and Engineering			Semester:	I
Subject: Web Essentials				
Subject Code:	26CS103	L – T – P - S:	42-0-0-48	
			Credits:	3

Sl. No	Course Objectives
	Understand the fundamental concepts of computers, hardware, software, and operating systems required for digital technology usage.
2	Understanding Linux fundamentals, file systems, permissions, and command-line concepts used in modern development environments.
3	Apply Git and GitHub concepts for version control, collaboration, and managing software projects.
4	Develop responsive web pages using HTML, CSS, and Tailwind CSS following modern web development practices.

Module	Description	Hrs
I	**Basics of Computers** Definition and working of computers, hardware and software concepts, input-process-output model, types of computers, CPU, RAM, storage, software types, and role of instructions in computing.	
II	**Linux Foundations** Operating system concepts, Linux environment, Linux architecture, kernel, shell, file system structure, terminal commands, file management, permissions, users, groups, environment variables, package management, Linux in web development environments.	
III	**Git, GitHub and Version Control** Need for version control, Git fundamentals, repositories, working directory, staging area, commits, history tracking, GitHub	

IV	collaboration, remote repositories, push, pull, clone operations.	

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Fundamentals of Web Development	Randy Connolly, Ricardo Hoar	3rd Edition, 2021 (Pearson)
2	HTML & CSS: Design and Build Websites	Jon Duckett	1st Edition, 2011 (Wiley)

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics	Jennifer Niederst Robbins	5th Edition, 2018
2	CSS: The Definitive Guide	Eric A. Meyer, Estelle Weyl	4th Edition, 2017
3	Pro Git	Scott Chacon, Ben Straub	2nd Edition, 2014
4	The Linux Command Line: A Complete Introduction	William E. Shotts Jr.	2nd Edition, 2019

Course Outcomes:

Course Outcomes	Descriptions
CO5	Develop responsive user interfaces using CSS and Tailwind CSS
	Explain fundamental computer concepts, including hardware,
CO1	
CO2	missions, and
CO3	l collaborative
	nts, semantic
CO1	markup, forms, and multimedia content.

Course Articulation Matrix

CO/PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	1	-	-	-	-	-	-	2	-

CO2	2	2	1	-	3	-	-	-	-	-	1	3	1
CO3	2	2	3	-	3	-	-	-	2	2	1	3	2
CO4	2	2	3	1	3	-	-	-	1	2	-	3	2
CO5	2	2	3	2	3	-	-	-	1	2	1	3	3

Department: Computer Science and Engineering			Semester:	I
Subject: Computer Fundamentals				
Subject Code:	26HS104	L – T – P - S:	56-0-0-64	
			Credits:	4

Sl. No.	Course Objectives
1	Explain the concepts and applications of computing
2	Explain data representation techniques and digital logic used in computing
3	Analyze computer networks, communication technologies, and systems, and software components.
4	Explain data representation techniques and digital logic used in computing

Module	Description	Hrs
1	Introduction to Computers and Computer Systems: Digital vs. analog computers.	12
2	Introduction to Computers and Computer Systems: Digital vs.	

II	analog computers; characteristics of computers; evolution from abacus to modern systems; generations of computers (1st–5th); classification: super, mainframe, mini, and micro computers; input and output devices. Computer memory: primary vs secondary memory; memory representation (bits, bytes, words); memory hierarchy (cache, RAM, ROM, storage); CPU registers; impact of cache on performance; programming language: a conceptual overview.	10
III	**Computational Thinking, Internet, and Digital Literacy Computationalthinking:** decomposition, pattern recognition, abstraction, algorithmic thinking. Introduction to algorithms: characteristics of a good algorithm. Flowcharts: symbols and design. Pseudocode. Problem-solving strategies. Programming paradigms overview (procedural, OOP, functional – conceptual). Software Development Lifecycle (SDLC – conceptual). Debugging mindset. Internet basics; web browsers and search engines; data vs information; introduction to databases; cloud storage basics; digital footprint and online safety; digital ethics and etiquette; productivity tools overview.	9
IV	**Emerging Technologies and Career Domains Overview:** the computing career landscape and why early awareness matters. Artificial Intelligence and Machine Learning: concepts and real-world applications. Data Science and Analytics. Computer Vision and NLP overview. Cloud Computing: concepts, service models, career roles. Cybersecurity: awareness, common threats, safety practices, career roles. Web Development: front-end vs back-end, career roles. Mobile App Development: concepts and distribution. UI/UX Design.	8
V	**Specialized Domains and Career Planning:** Robotics and IoT (concepts and careers). Blockchain and Web3 (conceptual overview). AR/VR and the Metaverse. DevOps and Site Reliability. Embedded Systems. Quantum Computing (conceptual glimpse). Tech Entrepreneurship and Product Management. Industry landscape: startups vs MNCs vs PSU.	6

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Computer Systems: A Programmer's Perspective	Randal E. Bryant, David R. O'Hallaron	3rd Edition, 2015
2	Fundamentals of Computers	E. Balagurusamy	8th Edition, 2023

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Computer Fundamentals	P. K. Sinha & Priti Sinha	6th Edition, 2013
2	Introduction to Computers	Peter Norton	7th Edition, 2018

Course Outcomes:

Course outcome	Descriptions
CO1	Students will explain the organization, components, characteristics, and evolution of computer systems.
CO2	Students will describe number systems, data representation techniques, and basic logic gates.
CO3	Students will explain data communication principles, network types, topologies, and internet fundamentals.
CO4	Students will apply computational thinking and design algorithms,
CO5	and draft a

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	2	1
CO2	3	2	-	-	1	-	-	-	-	-	-	2	1
CO3	2	2	-	-	2	1	-	1	-	1	-	2	2
CO4	2	3	2	1	2	-	-	-	-	1	-	3	2

CO5	1	1	-	-	2	2	-	2	1	2	3	2	3

Department: Computer Science and Engineering			Semester:	I
Subject: Python Programming Foundations Lab				
Subject Code:	26CS105	L – T – P - S:	0-0-28-2	
			Credits:	4

Sl. No	Course Objectives
1	Develop Python programs using variables, operators, control flow statements, loops, functions, and data structures through hands-on practice.
2	Apply problem-solving techniques to design, code, test, and debug Python programs for real-world problems.
3	Utilize object-oriented programming concepts, recursion, file handling, and exception handling to build working Python applications.

Module	Description	Hrs
I	**Python Basics and I/O Lab Practice** Programs using variables, data types, and arithmetic operators; Programs for input/output operations; Programs applying computational thinking using Python.	
II	**Control Flow and Iterative Problem Solving Lab** Programs using conditional branching (if, elif, else); Programs using looping structures (for, while); Programs using loop control statements (break, continue, pass); Programs solving pattern printing and nested loop problems	
III	**Functions, Strings and Collections Lab** Programs defining and using functions with parameters and scope; Programs on string manipulation, indexing, and slicing; Programs using Lists, Tuples, and Dictionaries; Programs applying Set operations to organize data	
IV	**Data Structures and Recursion Lab**	

	Programs on advanced list and dictionary operations; Programs implementing recursive algorithms (factorial, Fibonacci); Programs demonstrating base cases and recursive stack behaviour; Programs comparing iterative vs. recursive solutions	
V	**Object-Oriented Programming and File Handling Lab** Programs implementing classes, objects, and methods; Programs demonstrating inheritance, polymorphism, and encapsulation; Programs performing File I/O operations (read, write, append); Programs implementing exception handling in a mini-project	

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Think Python: How to Think Like a Computer Scientist	Allen B. Downey	3rd Edition, 2024
2	Automate the Boring Stuff with Python: Practical Programming for Total Beginners	Al Sweigart	2nd Edition, 2019

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Python Crash Course	Eric Matthes	3rd Edition, 2023
2	Head First Python	Paul Barry	2nd Edition, 2016
3	Learning Python	Mark Lutz	5th Edition, 2013

Course Outcomes:

Course outcome	Description
CO1	Implement recursive solutions and object-oriented programs using classes and OOP principles.
CO2	Build Python programs utilizing file handling and exception conditional
CO3	implement
CO4	

Course Articulation Matrix

CO/PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1													
CO2													
CO3													
CO4													

CO												
CO1	3				2			1			1	3	2
CO2	3	2	2		3							3	2
CO3	2	3	2		3							3	2
CO4	2	3	3		2							3	2
CO5	2	2	3		3							2	3

Department: Computer Science and Engineering			Semester:	I
Subject: Web Essentials Lab				
Subject Code:	26CS106	L – T – P - S:	0-0-28-2	
			Credits:	1

Sl. No	Course Objectives
	Build responsive web pages using HTML, CSS, and Tailwind CSS through hands-on exploration exercises.

Module	Description	Hrs
1	Basics of Computers Lab Exercises identifying hardware components and their functions; Hands-on exploration of operating system interfaces; Practical demonstrations of the input-process-output model; Exercises examining CPU, RAM, storage, and software types	
2	Practice Linux fundamentals, file systems, permissions, and command-line operations in a live terminal environment.	
3	Apply Git and GitHub commands for hands-on version control, collaboration, and managing software projects.	

II	**Linux Foundations Lab** Hands-on Linux terminal exercises for navigating the file system; Exercises on file management, permissions, users, and groups; Practice setting environment variables; Exercises using package managers to install and manage software	
IV	**Complete HTML Lab: Building the Structure of the Web** Hands-on exercises building HTML documents with tags, elements, and attributes; Exercises creating links, images, and lists; Practice writing semantic HTML, tables, and forms; Exercises adding metadata and applying accessibility and SEO basics	
V	**CSS Styling, Layout and Tailwind CSS Lab** Hands-on exercises using CSS selectors, cascade, inheritance, and the box model; Exercises building layouts with Flexbox and Grid; Practice with responsive design, typography, and animations; Exercises using Tailwind CSS utility classes and responsive utilities	

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Fundamentals of Web Development	Randy Connolly, Ricardo Hoar	3rd Edition, 2021 (Pearson)
2	HTML & CSS: Design and Build Websites	Jon Duckett	1st Edition, 2011 (Wiley)

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
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2	CSS: The Definitive Guide	Eric A. Meyer, Estelle Weyl	4th Edition, 2017
3	Pro Git	Scott Chacon, Ben Straub	2nd Edition, 2014

4	The Linux Command Line: A Complete Introduction	William E. Shotts Jr.	2nd Edition, 2019

Course Outcomes:

Course outcome	Descriptions
CO1	Demonstrate practical understanding of computer hardware,
CO2	sions, and
CO3	ntrol and
CO4	semantic

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	-	1	-	-	-	-	-	-	2	1
CO2	2	2	1	-	3	-	-	-	-	-	1	3	1
CO3	2	2	3	-	3	-	-	-	2	2	1	3	2
CO4	2	2	3	1	3	-	-	-	1	2	-	3	2
CO5	2	2	3	2	3	-	-	-	1	2	1	3	3

Develop responsive user interfaces using CSS and Tailwind techniques.

Department: Computer Science and Engineering		Semester:	I
Subject: Industrial Economics			
Subject Code:	26HS107	L – T – P - S:	28-0-0-32

			Credits:	**2**

Sl. No	Course Objectives
1	Analyze consumer behavior, price elasticity, and budget constraints using real-world tech and digital economy scenarios.
2	Evaluate short-run and long-run production, cost structures, and returns to scale in traditional and modern digital enterprises.
3	Differentiate market structures and strategic firm behavior, including monopolies and oligopolies in the tech and data sectors.
4	Analyze macroeconomic indicators, inflation, business cycles, and regulatory frameworks like GST and RBI policies on the industrial ecosystem.
5	

Module	Description	Hrs
I	**Economic Foundations for Industrial Economics :** What is Industrial Economics? Basic concepts of demand and supply: Law of demand and supply. Market equilibrium determination. Ceiling price and floor price.	6
II	**Consumer Behavior & Elasticity** Price elasticity of demand: Factors affecting price elasticity. Calculation. Relation between marginal revenue, demand and price elasticity. Income elasticity of demand and Cross elasticity of demand. Indifference curves. Budget Line.	6
III	**Production, Cost Analysis & Scale** Production and Cost analysis: Basic concepts. Production in the short-run and long-run. Cost analysis: Fixed, variable, average, and marginal costs. Finding the optimal combination of inputs. Returns to scale.	8
IV	**Market Structures & Strategic Decisions** Market: Characteristics of perfect competition. Profit maximisation in short-run and long-run. Firms with market power: Measurement and determinants of market power. Profit maximisation under monopoly: output and pricing decisions. Price discrimination, capturing consumer surplus. Strategic decision making in oligopoly markets.	4
V	**Macroeconomic Environment & Policy** National income: Concepts, Sources, Measurement, Difficulties, circular flow of income. Inflation: Cost-push and Demand-pull inflation, Effects and control of inflation. Business cycle. Functions of RBI. GST.	6

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition

1	Managerial Economics	D.N. Dwivedi	8th Ed., Vikas Publishing, 2018
2	Economics	Paul Samuelson & William Nordhaus	19th Ed., McGraw-Hill, 2010

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Industrial Organization: Theory and Practice	Don Waldman & Elizabeth Jensen	4th Ed., Routledge, 2013
2	Indian Economy	Ramesh Singh	Latest Ed., McGraw-Hill

Course Outcomes:

Course outcome	Descriptions
CO1	S… …emand-supply n… …rary tech and in…
CO2	S… …asticities and c… …strategies.
CO3	S… … and cost optimizations, identifying returns to scale in digital and physical industries.
CO4	Students will differentiate market structures, analyzing monopoly power, price discrimination, and oligopolistic strategies in today's digital economy.
CO5	Students will assess macroeconomic indicators, inflation trends, RBI actions, and GST impacts on industrial and startup growth.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	1	-	-	-	1	-	-	-	-	2	1	1
CO2	2	2	-	-	-	1	-	-	-	-	3	2	1
CO3	2	2	-	-	-	2	-	1	-	-	3	2	1
CO4	1	2	-	-	-	3	1	2	-	-	2	1	2
CO5	2	2	1	-	1	2	-	1	-	-	3	2	2

Syllabus for the Academic Year 2026-27